Algebra Mastery Pack — Practice Test B

Grade 11 algebra, functions & data analysis · 24 questions · suggested time 45 minutes · no calculator needed

Name: _____ Date: _____
Score: _____ / 24

Instructions. Write full working in the space under each question; answers alone do not show how you got there. Circle one letter for a multiple-choice question and write a single value on the answer line for a short-answer question. Answer every question — a blank scores zero. The answer key with a worked solution for every question is a separate PDF; mark your own paper only after you finish.

1. [Solving Linear Equations (for x)]

Solve $5x - 3 = 2x + 6$.

A) $x = 1$
B) $x = 2$
C) $x = 3$
D) $x = 9$

2. [Literal Equations & Solving for y]

Solve $C = 2\pi r$ for r.

A) $r = C \div (2\pi)$
B) $r = 2\pi C$
C) $r = C - 2\pi$
D) $r = 2\pi \div C$

3. [Linear Inequalities]

Solve $-2x > 8$.

A) $x < -4$
B) $x > -4$
C) $x < 4$
D) $x > 4$

4. [Systems of Linear Equations]

Solve by elimination: $x + y = 7$ and $x - y = 3$.

A) (5, 2)
B) (2, 5)
C) (5, 5)
D) (3, 4)

5. [Functions: Notation, Domain & Range]

If $g(x) = x^2 - 2$, what is $g(3)$?

A) 4
B) 7
C) 9
D) 11

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6. [Linear Functions: Slope, Intercepts & Equations]

In $y = 2x + 1$, what are the coordinates of the y-intercept?

A) (0, 1)
B) (1, 0)
C) (2, 1)
D) (0, 2)

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7. [Quadratic Functions & Equations]

Solve $x^2 - 9 = 0$ (difference of squares). Enter the positive solution.

Answer: _____

8. [Polynomials: Operations & Factoring]

Multiply: $(x + 2)(x + 3)$.

A) $x^2 + 5x + 6$
B) $x^2 + 6x + 5$
C) $x^2 + 5x + 5$
D) $x^2 + 6x + 6$

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9. [Radical & Rational Expressions]

Solve the rational equation $x/2 = 6$.

A) $x = 12$
B) $x = 3$
C) $x = 8$
D) $x = 6$

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10. [Exponential Functions & Growth/Decay]

$y = 5 \cdot (0.8)^x$ represents ___.

A) exponential decay
B) exponential growth
C) linear growth
D) no change

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11. [Data: Measures of Center & Spread]

Find the mode of {2, 4, 4, 6, 9}. Enter a number.

Answer: _____

12. [Scatter Plots, Line of Best Fit & Correlation]

A line of best fit ___.

A) approximates the overall trend of the data
B) connects every point
C) is always horizontal
D) must pass through the origin

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13. [Absolute Value Equations & Inequalities]

Solve $|x - 3| = 5$.

A) $x = 8$ only
B) $x = 8$ or $x = -2$
C) $x = 2$ only
D) $x = -8$ or $x = 2$

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14. [Arithmetic & Geometric Sequences]

An arithmetic sequence starts 3, 7, 11, … What is the 20th term?

Answer: _____

15. [Inverse Functions]

$f(x) = 3x - 6$. What is $f^{-1}(0)$?

Answer: _____

16. [Function Transformations & Vertex Form]

The graph of $y = f(x - 2)$ is the graph of $y = f(x)$...

A) shifted right 2 units
B) shifted left 2 units
C) shifted up 2 units
D) compressed horizontally

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17. [Systems of Linear Inequalities & Linear Programming]

Which point satisfies $x \geq 0$, $y \geq 0$ and $x + y \leq 4$?

A) (3, 2)
B) (−1, 2)
C) (4, 1)
D) (1, 1)

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18. [Probability from Two-Way Tables]

Survey of 200 students: how they travel and whether they arrived on time

	On time	Late	Total
Rides the bus	60	20	80
Walks	90	30	120
Total	150	50	200

What is P(late | walks), as a decimal?

Answer: _____

19. [Quadratic Functions & Equations]

Solve $x^2 + 2x - 8 = 0$ by factoring.

A) $x = -4$ or 2
B) $x = 4$ or -2
C) $x = -4$ or -2
D) $x = 4$ or 2

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20. [Polynomials: Operations & Factoring]

Factor $x^2 - 16$.

A) $(x + 4)(x - 4)$
B) $(x + 8)(x - 2)$
C) $(x - 8)(x + 2)$
D) $(x + 4)(x + 4)$

21. [Radical & Rational Expressions]

Solve $x/3 + 1 = 4$. Enter the value of x.

Answer: _____

22. [Exponential Functions & Growth/Decay]

$y = 2 \cdot (1.5)^x$ represents ___.

A) exponential growth
B) exponential decay
C) linear decay
D) no change

23. [Data: Measures of Center & Spread]

Find the range of {2, 4, 4, 6, 9}. Enter a number.

Answer: _____

24. [Scatter Plots, Line of Best Fit & Correlation]

A correlation coefficient r near 0 indicates ___.

A) weak or no linear correlation
B) strong positive correlation
C) strong negative correlation
D) a perfect line